

Magnetic degeneracy persists on a connected star

Disproof of conjecture 00000003721

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Take the connected simple graph $K_{1,3}$ with center 0, leaves 1, 2, 3 and edges $\{0, 1\}, \{0, 2\}, \{0, 3\}$. Assign arbitrary unit-modulus complex phases z_0, z_1, z_2 to these three oriented edges. In particular, $z_j \bar{z}_j = 1$ and $z_j \neq 0$. The magnetic adjacency matrix and the degree matrix are

$$A_z = \begin{pmatrix} 0 & z_0 & z_1 & z_2 \\ \bar{z}_0 & 0 & 0 & 0 \\ \bar{z}_1 & 0 & 0 & 0 \\ \bar{z}_2 & 0 & 0 & 0 \end{pmatrix}, \quad D = \text{diag}(3, 1, 1, 1).$$

The genuine Hermitian magnetic Laplacian is $L_z = D - A_z$. Consider

$$v_z = (0, \bar{z}_0, -\bar{z}_1, 0)^T, \quad w_z = (0, \bar{z}_0, 0, -\bar{z}_2)^T.$$

Direct multiplication gives $L_z v_z = v_z$ and $L_z w_z = w_z$. For example, the central coordinate of the first product is $-z_0 \bar{z}_0 + z_1 \bar{z}_1 = -1 + 1 = 0$; the leaf coordinates are the unchanged coordinates of v_z . The second calculation is identical with the third leaf replacing the second.

These eigenvectors are independent. If $av_z + bw_z = 0$, the coordinate at leaf 2 gives $-a\bar{z}_1 = 0$, hence $a = 0$; the coordinate at leaf 3 then gives $-b\bar{z}_2 = 0$, hence $b = 0$. Therefore

$$\dim_{\mathbb{C}} \ker(L_z - I) \geq 2 \quad \text{for every admissible phase tuple } z.$$

Eigenvalue one is thus degenerate for *all* phases. There is no phase at which the spectrum is nondegenerate, so degeneracy cannot disappear on any nonempty generic subset, whether generic is understood topologically, algebraically or in the almost-everywhere sense.

Geometric explanation. A tree has no magnetic cycle flux: its edge phases can be removed by a vertex gauge. Here the diagonal unitary $U = \text{diag}(1, \bar{z}_0, \bar{z}_1, \bar{z}_2)$ satisfies $L_z = UL_1U^*$. The ordinary star's repeated eigenvalue therefore persists. The explicit eigenvectors above already prove the required conclusion without invoking this general gauge principle.

Scope. The source states no restriction excluding trees or demanding cycle fluxes. The example is connected and has three actual, arbitrary edge phases; it is not an isolated-vertex or zero-operator example. It refutes the asserted generic disappearance of spectral degeneracy as written, independently of the additional unspecified flow-coprimality criterion. A statement restricted to some special class of graphs would need extra hypotheses.

Lean certificate. The project constructs the actual simple graph, proves connectedness and its degrees, builds the phased adjacency and verifies $L_z = D - A_z$ and Hermitian symmetry. It checks both eigenvectors, proves their linear independence in the genuine eigenvalue-one eigenspace and establishes its finite dimension is at least two for every unit-phase tuple.